

Midterm Exam

Introduction to Probability and Statistics

Name: _____

ID: _____

Time allotted: 75 minutes

For examiner's use only

Question	Points	Score
1	13	
2	10	
3	12	
4	15	
Total	50	

Instructions:

- Mobile phones and other electronic devices must be **turned off** and put away **before** the start of the exam. You are allowed to use a scientific calculator (**Not** the one on your smart phone. Sharing of calculators is **not** allowed as well).
- **No** food or drink, other than water, is allowed in the examination room.
- You are **not** allowed to communicate with other candidates.
- You are **not** allowed to leave your seat unless granted permission by an invigilator.
- Assume where necessary and **justify** your assumptions.
- You are required to fill **all** answers in the provided **answer booklet**.
- You are **allowed** to bring a cheat sheet. It must be a **handwritten** A4 sheet of paper. You may write on both sides of the sheet. Photocopied cheat sheets are **not** allowed.

Question 1:

You are visiting an office building for a job interview. The building has 250 workers.

To reduce the health effects of sitting all week, some employees play sports on weekends. Exactly 105 play cricket, 50 play football, 25 play table tennis, and the remaining employees play no sport. Assume these categories are mutually exclusive (no one plays more than one sport).

Among the cricket players, 30 have hypertension (high blood pressure). Among the football players, 8 have hypertension. Among the table tennis players, 12 have hypertension. Among the employees who play no sport, 60 have hypertension.

During your visit, you randomly stop a worker in the building and ask them whether they suffer from hypertension. If they answer in the positive, what is the probability they don't play any sport? **[13 points]**

Question 2:

Consider an experiment consisting of two consecutive rolls of a fair tetrahedral die (faces labeled 1, 2, 3, and 4). The events A , B and C are defined as:

- A : The first roll is even.
- B : The second roll is even.
- C : The sum of the two rolls is even.

- a) Test whether the events A , B , and C are independent. **[4 points]**
- b) Test whether the events A and B are independent. **[2 points]**
- c) Test whether the events A and B are independent when conditioned over C . **[4 points]**

Question 3:

PIN (Personal Identification Number) from a bank is a secret, numeric code that acts as your electronic signature to authenticate your identity during financial transactions, such as ATM withdrawals or point-of-sale purchases. PIN, is usually 4 to 6 digits long. The bank's current PIN selection policy is as follows:

- Length of 4 to 6 digits, i.e., user may select a 4 or a 5 or a 6-digit long PIN.
- May contain numbers (0–9)
- No spaces

- a) Under the current PIN selection policy, how many possible PINs (N_1) are there? **[4 points]**

Due to recent compromises and hacking attempts, a bank is considering that PIN selection should be made more robust against hacking and decoding by intelligent code-breaking machines. The Bank's cyber security guru has proposed the following criteria for the revised PIN selection policy:

- Length of 4 to 6 characters, i.e., user may select a 4 or a 5 or a 6 characters long PIN.
- Includes at least one special character out of the 5 allowed characters, *, #, @, &, !
- No spaces
- May contain numbers (0–9), lower- and upper-case letters (a–z, A–Z)
- Is case-sensitive.

- b) Under the revised PIN selection, how many possible PINs (N_2) are there? **[8 points]**

Question 4:

A university IT helpdesk receives support requests each hour. The random variable X represents the number of requests received in an hour and can take on 5 values, with the following probabilities:

$$P(X = 0) = 2p$$

$$P(X = 1) = 3p$$

$$P(X = 2) = 2.5p$$

$$P(X = 3) = 1.5p$$

$$P(X = 4) = p$$

- a) Calculate p and plot the probability mass function for the random variable X . [3 points]

The helpdesk has 2 staff members and each can handle at most 1 request per hour. If more than 2 requests arrive, the extra requests are left **unresolved**. Define a new random variable Y = number of unresolved requests in an hour.

- b) Write down and plot the probability mass function for the random variable Y . [4 points]
c) What is the probability that there are **unresolved** requests in an hour? [4 points]
d) Compute $E[Y]$, the expected number of **unresolved** requests per hour. [4 points]